

System Documentation

GrocerEase Inventory Management System

1. Introduction

GrocerEaseIMS is a PHP and MySQL inventory management system for grocery product control. It supports authenticated admin access, product records, categories, suppliers, product-supplier links, stock quantities, inventory movement logs, archived records, reporting charts, and database backup and recovery.

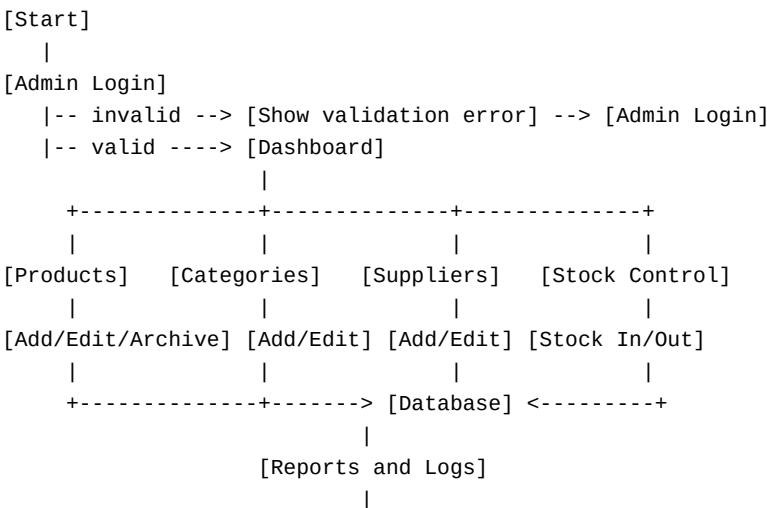
2. Objectives

- Maintain accurate product, supplier, category, stock, and movement data.
- Allow administrators to add, update, archive, restore, and permanently delete records.
- Provide dashboard visibility for active products, categories, suppliers, stock, low stock, and out-of-stock items.
- Protect system data through full SQL database backup and verified recovery.

3. Main Modules

- Authentication: validates admin login before allowing inventory access.
- Dashboard: displays KPIs, charts, inventory trends, and quick actions.
- Products and Categories: manages grocery product details and classification.
- Suppliers and Product Suppliers: tracks supplier information and preferred supplier cost.
- Stock Control: records stock in and stock out transactions.
- Archived Records: supports recovery or hard deletion of archived products, categories, and suppliers.
- Backup and Recovery: creates full SQL backups, validates restore files, and restores the database.

4. System Flowchart



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[Backup / Recovery]

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[End]

5. Database Tables

- users: administrator account details.
- categories: product category names, descriptions, status, and archive fields.
- products: product names, descriptions, category link, selling price, status, and archive fields.
- suppliers: supplier contact, address, company, and archive fields.
- product_supplier: product and supplier link, cost price, and preferred supplier flag.
- stocks: current stock quantity for each product.
- stock_movements: stock in/out transaction records and references.

6. Backup Strategy

The system uses full database backups. Each backup exports table structures and table data into a .sql file inside the backups folder. Full backup was selected because it is simple to demonstrate, easy to restore, and appropriate for the current system size.

7. Recovery Procedure

- Open Settings, then use the Backup and Recovery module.
- Select an existing backup from backup history or upload a .sql backup file.
- The system validates file type, readability, file size, and SQL contents.
- Before restore, the current database is automatically backed up as a safety backup.
- The SQL statements are executed with foreign key checks disabled during restore and enabled again after completion.
- SweetAlert feedback reports success, validation errors, or recovery errors.

8. Live Demonstration Script

- Show existing product, supplier, stock, or category records.
- Create a backup from Dashboard quick actions or Settings.
- Delete or modify selected records.
- Show that the records changed or disappeared.
- Restore the backup from Settings.
- Show that the original data returned.